STAT 801A Lab 6: Tests for Proportions

Insert Name

2026-10-12

t.test Function Review

Test	Use when	Call
One-sample	comparing one mean to a fixed value	`t.test(y, mu = mu0)`
Student's	two independent groups with same variance	`t.test(y ~ group, data = df, var.equal = TRUE)`
Welch's	two independent groups with different variances	`t.test(y ~ group, data = df)`
Paired	each value in one group has a partner in the other	`t.test(x1, x2, paired = TRUE)`

Important arguments:

- `mu`: the value in H_0. For one sample it is μ_0. For two samples it is the hypothesized difference. The default is 0.
- `alternative`: `"two.sided"` (the default), `"less"`, or `"greater"`.
- `var.equal`: `TRUE` gives Student's pooled test. The default, `FALSE`, gives Welch's test.
- `paired`: `TRUE` gives a paired test. Paired data must go in as two vectors in matching order.
- `conf.level`: the confidence level for the interval. The default is 0.95.

Student's and Welch's tests also accept two vectors in place of a formula, as in `t.test(x1, x2)`.

Testing a Single Proportion

So far the response has been numerical and summarized by a mean. When the response is categorical, we summarize it with a proportion instead.

A proportion is the fraction of a population with some characteristic, such as seeds that germinate or plants that are diseased. The population proportion is p. In a sample of size n with x successes, the sample proportion is $\hat{p} = x/n$.

To test $H_0 : p = p_0$, the test statistic is

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

This uses the normal distribution, so the sample has to be large enough: $np_0 \geq 10$ and $n(1 - p_0) \geq 10$.

Example: flower color

Mendel's laws predict that 3/4 of the offspring from a cross of two heterozygous pea plants will have purple flowers. A student makes this cross and grows 400 offspring. 288 have purple flowers.

Step 1: State the hypotheses

A departure in either direction would be evidence against the 3/4 ratio, so the test is two-tailed.

$$H_0 : p = 0.75 \qquad H_a : p \neq 0.75$$

Step 2: Check the validity conditions

```
n_pea <- 400
x_pea <- 288
p0_pea <- 0.75

n_pea * p0_pea
```

```
[1] 300
```

```r
n_pea * (1 - p0_pea)
```

```
[1] 100
```

We have 300 expected successes and 100 expected failures, both well above 10, so we can use the normal approximation.

Step 3: Compute the test statistic

```r
phat_pea <- x_pea / n_pea
z_pea <- (phat_pea - p0_pea) / sqrt(p0_pea * (1 - p0_pea) / n_pea)

phat_pea
```

```
[1] 0.72
```

```r
z_pea
```

```
[1] -1.385641
```

Find the rejection region at $\alpha = 0.05$:

```r
qnorm(0.025)
```

```
[1] -1.959964
```

```r
qnorm(0.975)
```

```
[1] 1.959964
```

We reject if $z < -1.96$ or $z > 1.96$. Our z of -1.39 falls between them, so we fail to reject H_0.

Step 4: Compute the p-value

```r
2 * pnorm(abs(z_pea), lower.tail = FALSE)
```

```
[1] 0.1658567
```

Step 5: Confirm with prop.test

```r
prop.test(x = 288, n = 400, p = 0.75, correct = FALSE)
```

```
    1-sample proportions test without continuity correction

data:  288 out of 400, null probability 0.75
X-squared = 1.92, df = 1, p-value = 0.1659
alternative hypothesis: true p is not equal to 0.75
95 percent confidence interval:
 0.6740660 0.7617486
sample estimates:
   p
0.72
```

`correct = FALSE` turns off a continuity correction that R adds by default. The number of successes follows a binomial distribution, which is discrete, and the correction adjusts the count by 0.5 so the continuous normal distribution approximates it more closely. Without it, the output will match the hand calculation. With small samples, it is typically better to leave it on.

R reports `X-squared` instead of z. It equals z^2, here $(-1.386)^2 = 1.92$, and the p-value is the same.

For a one-tailed test, add `alternative = "less"` or `alternative = "greater"`, the same as `t.test`.

Note: The confidence interval from `prop.test` uses a different method than $\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$, so it will be close to a hand calculation but not exact. R's interval always stays between 0 and 1, which matters when $\hat{p}$ is near 0 or 1, since a proportion can't be below 0 or above 1.

Step 6: Write the conclusion

There is not enough evidence that the proportion of purple-flowered offspring differs from 3/4 ($z = -1.39$, $p = 0.166$). In the sample, 72% had purple flowers, with a 95% confidence interval of 67.4% to 76.2%.

Comparing Two Proportions

To compare two independent groups on a yes/no outcome, test whether the two population proportions are equal.

$$H_0 : p_1 = p_2 \qquad H_a : p_1 \neq p_2$$

Under H_0 both samples estimate the same p, so they are pooled. This is the same idea as the pooled variance for means.

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} \qquad z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Each group needs at least 10 successes and 10 failures.

Example: website layouts

An online store tests a redesigned product page. Visitors are randomly shown either the old page or the new one, 200 visitors each. On the old page, 34 visitors make a purchase. On the new page, 52 do.

Step 1: State the hypotheses

Group 1 is the old page and group 2 is the new page.

$$H_0 : p_1 = p_2 \qquad H_a : p_1 \neq p_2$$

Step 2: Check the validity conditions

Enter the counts as vectors, group 1 first.

```
x_web <- c(34, 52)
n_web <- c(200, 200)

x_web
```

```
[1] 34 52
```

```
n_web - x_web
```

```
[1] 166 148
```

The smallest of the four counts is 34, so the conditions hold.

Step 3: Compute the test statistic

```
phat1_web <- 34/200
phat2_web <- 52/200
phat_web <- (34+52) / (200+200)

z_web <- (phat1_web - phat2_web) /
  sqrt(phat_web * (1 - phat_web) * (1 / 200 + 1 / 200))

phat1_web
```

```
[1] 0.17
```

```
phat2_web
```

```
[1] 0.26
```

```
z_web
```

```
[1] -2.190728
```

Find the rejection region at $\alpha = 0.05$:

```
qnorm(0.025)
```

```
[1] -1.959964
```

```
qnorm(0.975)
```

```
[1] 1.959964
```

We reject if $z < -1.96$ or $z > 1.96$. Our z of -2.19 is below -1.96, so we reject H_0.

Step 4: Compute the p-value

```
2 * pnorm(abs(z_web), lower.tail = FALSE)
```

```
[1] 0.02847148
```

Step 5: Confirm with prop.test

```
prop.test(x = c(34, 52), n = c(200, 200), correct = FALSE)
```

```
    2-sample test for equality of proportions without continuity correction

data:  c(34, 52) out of c(200, 200)
X-squared = 4.7993, df = 1, p-value = 0.02847
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.170035196 -0.009964804
sample estimates:
prop 1 prop 2
  0.17   0.26
```

R subtracts group 2 from group 1, and `X-squared` is again z^2.

Step 6: Write the conclusion

There is evidence that the purchase rate differs between the two pages ($z = -2.19$, $p = 0.028$). On the old page, 17% of visitors made a purchase, compared with 26% on the new page. We are 95% confident that the new page's purchase rate is between 1 and 17 percentage points higher.

When we don't have enough successes and failures for the normal approximation

If the conditions fail, the normal approximation is not reliable, so use an exact test instead. For one proportion, `binom.test(x, n, p)` gets the p-value directly from the binomial distribution. For two proportions, `fisher.test` does the same job. It takes a table with one row per group, with successes and failures in the columns:

```
web_table <- rbind(old = c(purchase = 34, no_purchase = 166),
                   new = c(purchase = 52, no_purchase = 148))
web_table
```

```
    purchase no_purchase
old       34         166
new       52         148
```

```
fisher.test(web_table)
```

```

    Fisher's Exact Test for Count Data

data:  web_table
p-value = 0.03815
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.3468322 0.9728492
sample estimates:
odds ratio
 0.5837448
```

`fisher.test` reports an odds ratio instead of a difference in proportions (the confidence interval is also for an odds ratio). Its p-value, 0.038, is a little larger than the 0.028 from `prop.test`, which is typical of exact tests.

We still compare the p-value to $\alpha = 0.05$. Because counts move in whole-number steps, an exact test's actual Type I error rate is at or below α rather than exactly α.

Practice Problems

Where a problem asks for hypotheses, write them in LaTeX.

Problem 1: Choosing a t Test

For each scenario, decide which version of `t.test` fits, run it, and write a conclusion in context at $\alpha = 0.05$. For two independent groups, use the rule of thumb from Lab 5: if the larger standard deviation is more than twice the smaller, use Welch's test.

Scenario A

A seed company compares ear length (cm) of a hybrid corn variety with an open-pollinated variety. Ten ears of each are measured.

Run this chunk to create the data.

```
ears <- data.frame(
  variety = rep(c("hybrid", "open"), each = 10),
  length = c(19.8, 20.4, 19.5, 20.9, 20.1, 19.7, 20.6, 20.2, 19.9, 20.3,
             17.2, 21.5, 15.8, 19.9, 22.6, 16.4, 18.8, 21.1, 15.5, 18.3))
```

(a) Which version of `t.test` fits? Explain.

```
sd(ears$length[ears$variety == "hybrid"])
```

```
[1] 0.4299871
```

```
sd(ears$length[ears$variety == "open"])
```

```
[1] 2.507965
```

Welch's. These are two independent groups, and the open-pollinated standard deviation (2.51) is almost six times the hybrid's (0.43).

(b) Run the test. Report t, df, and the p-value.

```r
t.test(length ~ variety, data = ears)
```

```
    Welch Two Sample t-test

data:  length by variety
t = 1.7771, df = 9.5286, p-value = 0.1074
alternative hypothesis: true difference in means between group hybrid and group open is not e
95 percent confidence interval:
 -0.374989  3.234989
sample estimates:
mean in group hybrid   mean in group open
               20.14                18.71
```

(c) Write a conclusion in context.

There is not enough evidence that mean ear length differs between the two varieties ($t = 1.78$, $df = 9.53$, $p = 0.107$). The hybrid ears averaged 1.4 cm longer, but the open-pollinated ears varied too much to rule out no difference.

Scenario B

The built-in `sleep` data records the extra hours of sleep 10 patients got on each of two drugs. Every patient took both drugs. `ID` identifies the patient and `group` identifies the drug.

```r
data(sleep)
sleep
```

```
   extra group ID
1    0.7     1  1
2   -1.6     1  2
3   -0.2     1  3
4   -1.2     1  4
5   -0.1     1  5
6    3.4     1  6
7    3.7     1  7
8    0.8     1  8
9    0.0     1  9
10   2.0     1 10
11   1.9     2  1
```

```
12   0.8     2  2
13   1.1     2  3
14   0.1     2  4
15  -0.1     2  5
16   4.4     2  6
17   5.5     2  7
18   1.6     2  8
19   4.6     2  9
20   3.4     2 10
```

(a) Which version of `t.test` fits? Explain.

Paired. Each patient appears once under each drug, so the two measurements for a patient form a pair. The rows are sorted by ID within each drug, so the vectors in part (b) line up patient by patient.

(b) Run the test. Report t, df, and the p-value.

```
drug1 <- sleep$extra[sleep$group == 1]
drug2 <- sleep$extra[sleep$group == 2]

t.test(drug1, drug2, paired = TRUE)
```

```
    Paired t-test

data:  drug1 and drug2
t = -4.0621, df = 9, p-value = 0.002833
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -2.4598858 -0.7001142
sample estimates:
mean difference
          -1.58
```

(c) Write a conclusion in context.

There is strong evidence that the two drugs differ in mean extra sleep ($t = -4.06$, $df = 9$, $p = 0.003$). Patients slept an average of 1.58 hours more on drug 2 than on drug 1

Scenario C

Bags of feed from a local mill are labeled 50 lb. A buyer suspects the bags are underfilled and weighs 12 of them.

Run this chunk to create the data.

```
bags <- c(49.6, 50.2, 49.1, 49.8, 50.4, 49.3, 49.7, 49.0, 50.1, 49.5, 49.4, 49.9)
```

(a) Which version of `t.test` fits? Explain.

One-sample, since a single mean is compared to the labeled 50 lb. The buyer suspects underfilling, so the test is lower-tailed.

(b) Run the test. Report t, df, and the p-value.

```
t.test(bags, mu = 50, alternative = "less")
```

```
    One Sample t-test

data:  bags
t = -2.6512, df = 11, p-value = 0.01127
alternative hypothesis: true mean is less than 50
95 percent confidence interval:
     -Inf 49.89246
sample estimates:
mean of x
 49.66667
```

(c) Write a conclusion in context.

There is evidence that the mean bag weight is below the labeled 50 lb ($t = -2.65$, $df = 11$, $p = 0.011$). The 12 bags averaged 49.67 lb.

Scenario D

Twenty pigs in individual pens are randomly assigned to one of two diets, 10 per diet. Weight gain (lb) over four weeks is recorded.

Run this chunk to create the data.

```r
pigs <- data.frame(
  diet = rep(c("A", "B"), each = 10),
  gain = c(52.4, 55.8, 49.6, 57.3, 53.1, 50.8, 54.9, 51.7, 56.2, 53.6,
           56.9, 59.4, 54.2, 61.8, 57.5, 55.1, 60.3, 58.6, 62.4, 56.8))
```

(a) Which version of `t.test` fits? Explain.

```r
sd(pigs$gain[pigs$diet == "A"])
```

```
[1] 2.496308
```

```r
sd(pigs$gain[pigs$diet == "B"])
```

```
[1] 2.713342
```

Student's. These are two independent groups with similar standard deviations, so pooling is reasonable.

(b) Run the test. Report t, df, and the p-value.

```r
t.test(gain ~ diet, data = pigs, var.equal = TRUE)
```

```
    Two Sample t-test

data:  gain by diet
t = -4.0826, df = 18, p-value = 0.0006988
alternative hypothesis: true difference in means between group A and group B is not equal to
95 percent confidence interval:
 -7.209515 -2.310485
sample estimates:
mean in group A mean in group B
          53.54           58.30
```

(c) Write a conclusion in context.

There is strong evidence that mean weight gain differs between the two diets ($t = -4.08$, $df = 18$, $p < 0.001$). Pigs on diet B gained an estimated 4.76 lb more over four weeks (95% CI: 2.31 to 7.21 lb).

Problem 2

A seed company advertises a 90% germination rate for its alfalfa seed. A buyer suspects the true rate is lower. She plants 200 seeds and 171 germinate.

(a) Compute the sample proportion.

```
n_seed <- 200
x_seed <- 171
p0_seed <- 0.90

phat_seed <- x_seed / n_seed
phat_seed
```

```
[1] 0.855
```

(b) State the hypotheses.

$$H_0 : p = 0.90 \qquad H_a : p < 0.90$$

(c) Are the validity conditions to use the normal approximation met here?

```
n_seed * p0_seed
```

```
[1] 180
```

```
n_seed * (1 - p0_seed)
```

```
[1] 20
```

Yes. $np_0 = 180$ and $n(1 - p_0) = 20$, both at least 10.

(d) Compute the test statistic and p-value by hand.

```
z_seed <- (phat_seed - p0_seed) / sqrt(p0_seed * (1 - p0_seed) / n_seed)
z_seed
```

```
[1] -2.12132
```

```r
pnorm(z_seed)
```

```
[1] 0.01694743
```

$z = -2.12$ and $p = 0.017$. For a lower-tailed test, the p-value is the area to the left of z.

(e) Confirm with `prop.test`. Remember `correct = FALSE` and the alternative.

```r
prop.test(x = 171, n = 200, p = 0.90, alternative = "less", correct = FALSE)
```

```
    1-sample proportions test without continuity correction

data:  171 out of 200, null probability 0.9
X-squared = 4.5, df = 1, p-value = 0.01695
alternative hypothesis: true p is less than 0.9
95 percent confidence interval:
 0.000000 0.891215
sample estimates:
    p
0.855
```

`X-squared` is 4.5, which is $(-2.12)^2$, and the p-value of 0.017 matches part (d).

(f) Write a conclusion in context at $\alpha = 0.05$.

There is evidence that the germination rate is below the advertised 90% ($z = -2.12$, $p = 0.017$). Of the 200 seeds tested, 85.5% germinated.

Problem 3

Waterhemp is a common weed in Nebraska crop fields, and some populations are resistant to glyphosate. A weed scientist collects plants from fields in two counties and tests each one. In County A, 64 of 110 plants are resistant. In County B, 45 of 115 are.

(a) Compute each sample proportion.

```r
x_weed <- c(64, 45)
n_weed <- c(110, 115)

x_weed / n_weed
```

```
[1] 0.5818182 0.3913043
```

(b) State the hypotheses. Let group 1 be County A.

$$H_0 : p_1 = p_2 \qquad H_a : p_1 \neq p_2$$

(c) Are the validity conditions to use the normal approximation met here?

```r
x_weed
```

```
[1] 64 45
```

```r
n_weed - x_weed
```

```
[1] 46 70
```

Yes. The smallest of the four counts is 45, so the conditions hold.

(d) Run `prop.test`. Report the p-value.

```r
prop.test(x = x_weed, n = n_weed, correct = FALSE)
```

```
    2-sample test for equality of proportions without continuity correction

data:  x_weed out of n_weed
X-squared = 8.1704, df = 1, p-value = 0.004258
alternative hypothesis: two.sided
95 percent confidence interval:
 0.06224401 0.31878366
sample estimates:
   prop 1    prop 2
0.5818182 0.3913043
```

(e) Interpret the 95% confidence interval.

We are 95% confident that the proportion of resistant plants is between 6.2 and 31.9 percentage points higher in County A than in County B.

(f) Write a conclusion in context.

There is strong evidence that the rate of glyphosate resistance differs between the two counties ($z = 2.86$, $p = 0.004$). In the sample, 58.2% of County A plants were resistant, compared with 39.1% of County B plants.

Final Step

Render your document. Click the Render button at the top of the editor, or press Ctrl + Shift + K.

If the render fails, read the error message. It names the chunk that caused the problem. The most common causes are a missing closing parenthesis, or an object used before it was created.